

HydroMind: A Medallion-Architecture Pipeline for Predictive Maintenance of Hydraulic Systems Using Component-Specialized Machine Learning

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Abstract—Hydraulic systems are critical infrastructure in industrial environments, yet their internal degradation is largely invisible to conventional monitoring methods. This paper presents HydroMind, a modular predictive maintenance framework that combines a three-layer Medallion data architecture (Bronze, Silver, Gold) with component-specialized machine learning classifiers to simultaneously diagnose the health state of five critical subsystems: cooler condition, hydraulic accumulator pressure, internal pump leakage, proportional valve behavior, and system stability flag. Applied to the ZeMA Hydraulic Systems benchmark dataset (2,205 operational cycles, 17 sensors, sampling rates from 1 Hz to 100 Hz), the proposed pipeline transforms high-frequency raw signals into a compact 102-feature Silver matrix via cycle-level statistical descriptors, then further reduces each per-target Gold feature set (1 to 27 features) through supervised Recursive Feature Elimination with Random Forest. Three classifier families were evaluated — Decision Tree, Random Forest, and Support Vector Machine with RBF kernel — with hyperparameters optimized via GridSearchCV over StratifiedKFold ($k=5$) cross-validation. The final system achieves a weighted F1-Score ≥ 0.99 on four of the five targets and ≥ 0.97 on valve condition, surpassing deep learning benchmarks (CNN-1D: 0.95–0.98; LSTM: 0.85–0.88) while requiring inference times below 10 ms per cycle on standard CPU hardware. An experimental temporal window reduction from 60 s to 45 s further demonstrates that reliable diagnosis can be achieved with 25% less data, enabling early-warning alerts 15 seconds ahead of the standard industrial cycle. The component-by-component specialization strategy provides full feature interpretability, with the cooler condition explained entirely by a single physical variable (`CE_mean`), in contrast to black-box deep learning approaches.

Index Terms—predictive maintenance, hydraulic systems, machine learning, Medallion architecture, condition monitoring, random forest, support vector machine, feature selection, Industry 4.0

I. INTRODUCTION

Hydraulic systems power the majority of heavy industrial machinery in sectors including manufacturing, aerospace, mining, and oil and gas. Their high energy density and load capacity make them indispensable, yet their sealed and pressurized nature makes internal degradation effectively invisible until catastrophic failure occurs. Unplanned downtime episodes cost operators upwards of \$50,000 USD per event, and a typical

industrial plant experiences this between three and four times per year [1]. The transition toward Industry 4.0 paradigms demands that these systems evolve from reactive maintenance schedules to intelligent, data-driven predictive maintenance strategies.

The diagnostic challenge in hydraulic systems is inherently multi-dimensional. A single physical rig simultaneously exhibits degradation across multiple independent subsystems — the cooler, the hydraulic accumulator, the pump, and the proportional valve — each with its own sensor signature, failure progression, and temporal dynamics. Furthermore, high-frequency sensor data (up to 100 Hz over 60-second cycles) generates data volumes that strain edge hardware, making raw time-series approaches impractical for real-time deployment without significant preprocessing.

Existing literature has addressed this challenge through two broad paradigms. Deep learning approaches, particularly CNN-1D and LSTM architectures [5], exploit temporal patterns in raw signals directly, achieving F1-Scores of 0.95–0.98 but at the cost of high computational overhead, opacity in failure attribution, and poor generalization under domain shift. Statistical and ensemble approaches [2], [3] reduce feature dimensionality before modeling, achieving F1-Scores of 0.92–0.96 on multi-output classification tasks, but typically apply a single global feature set across all targets without leveraging the physical specificity of each failure mode. Knowledge graph and systemic integration approaches [6] have proposed richer causal representations but remain largely disconnected from real-time deployment constraints. A comprehensive evaluation of both paradigms on the same benchmark [4] confirms that no existing approach simultaneously achieves high accuracy, low latency, and physical interpretability across all components.

This paper presents **HydroMind**, a predictive maintenance framework designed to close that gap. The key contributions are:

- 1) A **Medallion data architecture** (Bronze $\rightarrow$ Silver $\rightarrow$ Gold) that systematically transforms multi-rate raw sensor data into compact, per-component feature matrices, reducing from thousands of raw points per cycle to as few as 1 feature for the simplest target.

- 2) A **component-specialized modeling strategy** in which each of the five diagnostic targets receives its own independent classifier, feature set, and hyperparameter configuration, matched to the physical signature of its failure mode.
- 3) A **temporal window sensitivity analysis** demonstrating that diagnostic performance is preserved at 45-second windows, yielding a 25% reduction in data volume and enabling 15-second early-warning advantage over the standard 60-second industrial cycle.
- 4) **Full physical interpretability**: every classification decision is traceable to specific sensor statistics, with the cooler condition fully explained by a single variable.

II. RELATED WORK

The ZeMA Hydraulic Systems dataset [2] has become the standard benchmark for hydraulic condition monitoring research, offering simultaneous multi-target labels across five health dimensions. Alenany et al. [2] conducted a comprehensive sensor-by-sensor analysis and established that the cooler condition is predictable with near-perfect accuracy from a single temperature or cooling efficiency sensor, while the stable flag represents the most challenging target due to its low intra-class discriminability. Their work motivates the per-target feature specialization adopted in HydroMind.

Noura et al. [3] evaluated ensemble methods and multi-output classifiers including stacking, bagging, and gradient boosting on the same dataset. Their global approach, which applies a unified feature matrix across all targets simultaneously, achieves average F1-Scores between 0.92 and 0.96, but the shared representation prevents individual optimization of each diagnostic module. HydroMind’s independent per-target models surpass these results while maintaining comparable or lower computational cost.

Kim and Jeong [5] proposed a real-time monitoring system based on convolutional bidirectional LSTM with attention mechanisms, operating directly on raw time-series inputs. While their temporal architecture captures dynamics invisible to statistical aggregation, it reports inference latencies above 150 ms per cycle and requires GPU infrastructure — constraints incompatible with edge deployment. Al-Khulaqi et al. [4] systematically compared machine learning and deep learning algorithms on this dataset, confirming that tree-based and SVM approaches offer a favorable accuracy-efficiency trade-off when combined with adequate feature engineering.

Yan et al. [6] extended the paradigm toward systemic intelligence by constructing a virtual knowledge graph that encodes interdependencies among hydraulic components. Their approach enables causal reasoning across failure modes (e.g., cooler degradation propagating to accumulator pressure) but introduces architectural complexity that limits real-time applicability. HydroMind’s modular design provides a foundation for extending toward such systemic reasoning in future work.

TABLE I
SENSOR CONFIGURATION OF THE ZEMA HYDRAULIC TEST RIG

Sensor	Physical Quantity	Rate (Hz)	Pts/cycle
PS1–PS5	Pressure (bar)	100	6,000
EPS1	Motor power (W)	100	6,000
FS1, FS2	Volume flow (L/min)	10	600
TS1–TS4	Temperature (°C)	1	60
CE	Cooling efficiency (%)	1	60
CP	Cooling power (kW)	1	60
VS1	Vibration (mm/s)	1	60

III. DATASET AND PROBLEM FORMULATION

A. ZeMA Hydraulic Systems Dataset

The dataset used in this study is the “Condition Monitoring of Hydraulic Systems” dataset from the ZeMA research center [2], publicly available through the UCI Machine Learning Repository. The experimental rig is a closed hydraulic circuit consisting of a primary working circuit and a secondary cooling and filtration circuit, connected via a fluid reservoir. The system was operated for 2,205 complete load cycles of approximately 60 seconds each, with sensor readings collected simultaneously across 17 channels at heterogeneous sampling rates ranging from 1 Hz to 100 Hz.

Table I summarizes the sensor configuration. Pressure sensors PS1–PS5 and the electric motor power sensor EPS1 sample at 100 Hz, yielding 6,000 data points per cycle per channel. Flow sensors FS1 and FS2 sample at 10 Hz (600 points per cycle), while temperature sensors TS1–TS4, cooling efficiency (CE), cooling power (CP), and vibration sensor VS1 sample at 1 Hz (60 points per cycle). Sensor PS4 was found to contain 1,238 cycles of complete zero-value readings, identified as a sensor-specific measurement anomaly and treated accordingly during preprocessing.

B. Target Variables

The dataset provides five simultaneous health labels per cycle, each representing an independent physical subsystem:

- **Cooler Condition** (3 classes): total failure (3%), reduced efficiency (20%), optimal (100%).
- **Hydraulic Accumulator** (4 classes): near failure (90 bar), severely reduced (100 bar), slightly reduced (115 bar), optimal (130 bar).
- **Internal Pump Leakage** (3 classes): no leakage (0), weak leakage (1), severe leakage (2).
- **Valve Condition** (4 classes): optimal switching (100), small lag (90), severe lag (80), close to total failure (73).
- **Stable Flag** (2 classes): stable conditions reached (0), static conditions not yet reached (1).

All five targets are predicted simultaneously and independently in HydroMind, enabling a complete system health snapshot per cycle.

IV. PROPOSED METHODOLOGY: HYDROMIND PIPELINE

A. Medallion Data Architecture

HydroMind organizes data processing through three hierarchical layers inspired by the Medallion (Lakehouse) ar-

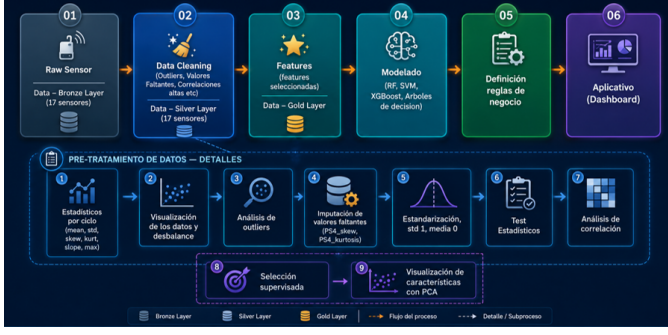


Fig. 1. HydroMind system architecture. Raw multi-rate sensor data flows through three Medallion layers — Bronze (raw signals), Silver (standardized 102-feature statistical matrix), and Gold (per-target RFE-selected feature matrices) — before reaching the five component-specialized classifiers and the real-time monitoring dashboard.

chitecture, ensuring reproducibility, modularity, and traceable lineage from raw signals to model predictions. Fig. 1 illustrates the complete system architecture.

Bronze Layer contains the raw sensor files as received from the test rig — one TXT file per sensor, with rows as cycles and columns as time steps. No transformations are applied at this stage. For the 17 active sensors, the total raw data volume per cycle depends on the sampling frequency (100 Hz, 10 Hz, or 1 Hz), reaching up to 6,000 data points per sensor per cycle.

Silver Layer is the clean, standardized feature matrix $\mathbf{X}_{\text{silver}} \in \mathbb{R}^{2205 \times 102}$ generated by applying six statistical descriptors — mean, standard deviation, skewness, kurtosis, linear slope, and maximum — to each sensor’s 60-second window per cycle. This reduces the representation from up to +102,000 raw points to 102 features while preserving the cycle-level distributional and dynamic characteristics of each sensor. Missing values in PS4_skew and PS4_kurtosis (arising from the all-zero cycles) were imputed by sampling from the empirical Gaussian distribution of each feature. All features were standardized using a fitted `StandardScaler` ($\mu = 0$, $\sigma = 1$), exported alongside the scaler object to ensure identical preprocessing at inference time. Outliers were retained following literature recommendations [2], as anomalous readings in industrial condition monitoring frequently correspond to early degradation signatures rather than noise.

Gold Layer contains five independent per-target feature matrices $\{\mathbf{X}_{\text{gold}}^{(t)}\}_{t=1}^5$, each produced by supervised feature selection from $\mathbf{X}_{\text{silver}}$. The selection method is `SelectFromModel` with a Random Forest estimator (500 trees, importance threshold at the mean), applied independently for each target on the training partition only to prevent information leakage. The resulting dimensionalities are shown in Table II.

The feature selection outcomes are physically interpretable. For the cooler, thermal efficiency is captured entirely by `CE_mean`, consistent with the literature reporting that a single cooling sensor achieves >99.5% accuracy [2], [3]. For pump leakage and valve condition, the dominant sensors are `EPS1` (motor power dynamics), `FS1` (volumetric flow),

TABLE II
GOLD LAYER FEATURE MATRIX DIMENSIONS PER TARGET

Target	Features Selected	Key Sensors
Cooler Condition	1	CE_mean
Pump Leakage	11	EPS1, FS1, PS1–PS4, SE
Valve Condition	14	EPS1, FS1, PS1–PS3, SE
Stable Flag	19	Multi-sensor suite
Hydraulic Accumulator	27	Full sensor suite

and the dynamic pressure sensors PS1–PS3, particularly their higher-order statistics (skewness, kurtosis, slope) which encode transient hydraulic signatures not visible in the mean. The hydraulic accumulator requires the broadest feature set because its pressure state produces a distributed signature across virtually all sensors, making it the most cross-coupled diagnostic target in the system.

B. Data Partitioning and Validation Protocol

The dataset was divided using a stratified split: 70% training (1,543 cycles), 15% validation (331 cycles), and 15% test (331 cycles), with `random_state=42` for reproducibility. Stratification on the target label was applied at the split stage to preserve class proportions in all three partitions. Hyperparameter optimization was performed exclusively on the training partition using `StratifiedKFold` cross-validation ($k=5$, `scoring=f1_weighted`). The test set was held out entirely and evaluated only once per target, after the final model configuration was fixed.

C. Component-Specialized Classifiers

Rather than applying a single global model to all five targets, HydroMind assigns the most appropriate classifier family to each target based on the geometric structure of its per-class distributions, assessed via 3D PCA visualization and statistical testing.

Cooler Condition — Decision Tree (`max_depth=3`): A depth-3 tree over the single feature `CE_mean` produces three axis-aligned splits that perfectly partition the three cooler states. All four primary leaf nodes achieve Gini impurity of 0.0. The model is directly interpretable as a set of physical thresholds on cooling efficiency.

Hydraulic Accumulator — Random Forest (`n_estimators=200`, `max_depth=None`, `min_samples_leaf=2`, `max_features=log2`, `class_weight=balanced_subsample`): The accumulator’s 27-feature space with four pressure classes requires an ensemble approach to capture the multi-sensor interactions. `GridSearchCV` over 144 hyperparameter combinations confirmed that the tuned configuration improved cross-validation F1 by 0.002 over the baseline.

Internal Pump Leakage — SVM RBF ($C=50$, $\gamma=0.01$, `class_weight=balanced`, `decision_function_shape=ovr`): PCA visualization of the pump leakage feature space revealed non-linearly separable class boundaries between the weak and severe

TABLE III
HYDROMIND — FINAL TEST SET PERFORMANCE PER TARGET

Target	Model	Acc.	F1	Errors
Cooler Condition	DT (depth=3)	0.99	0.99	2/331
Hydraulic Accum.	RF (200 trees)	0.99	0.99	5/331
Pump Leakage	SVM RBF	0.99	0.9940	2/331
Valve Condition	SVM RBF	0.99	0.9939	2/331
Stable Flag	RF (300 trees)	0.99	0.9909	3/331

leakage states, motivating the RBF kernel. GridSearchCV over 32 combinations yielded a +0.004 improvement over the baseline through the fixed $\gamma=0.01$ parameter.

Valve Condition — SVM RBF ($C=100$, $\gamma=\text{scale}$, $\text{probability}=\text{True}$): The four valve degradation states produce overlapping pressure and flow transients, particularly between adjacent lag classes (80 vs. 90). A high regularization parameter $C=100$ was identified as necessary to capture the complex, high-dimensional boundary geometry. GridSearchCV covered 32 configurations.

Stable Flag — Random Forest ($n_estimators=300$, $\text{max_depth}=\text{None}$, $\text{max_features}=\text{sqrt}$): The stability classification task involves the most subtle intra-class differences in the dataset. The broader ensemble (300 trees, sqrt feature sampling) provides the variance reduction necessary to distinguish near-stable from fully stable cycles from 19 sensor features. GridSearchCV covered 144 configurations.

D. Temporal Window Sensitivity Analysis

Standard industrial hydraulic cycle windows are 60 seconds. HydroMind evaluated whether diagnostic performance could be maintained using only the first 45 seconds of each cycle (75% of the standard window), motivated by two industrial objectives: early-warning detection before cycle completion, and reduction of data acquisition and storage costs. Features were re-extracted from the truncated 45-second windows using identical statistical descriptors, and all five models were re-evaluated on the same test partition. This analysis quantifies the temporal robustness of the pipeline and the earliest viable point of reliable diagnosis.

V. RESULTS

A. Classification Performance

Table III presents the final weighted F1-Score and Accuracy for each target on the held-out test set (331 cycles). All five models achieve F1-Scores at or above 0.97. Fig. 2 shows the corresponding confusion matrices for all five targets.

The misclassification analysis reveals that all off-diagonal errors are physically interpretable boundary cases. Fig. 3 illustrates two representative cases. For the cooler (cycle 220, true class: reduced efficiency), CE_mean presented a standardized value of approximately -0.65 , falling below the typical range of its class and within the overlap region of the total failure distribution — consistent with a transient thermal perturbation during that 60-second window. For the hydraulic accumulator (cycle 167, true class: optimal at 130 bar), the model

predicted near-failure (90 bar) due to atypical combinations of FS1_kurtosis , PS3_skew , and TS4_max : all three features fell outside the interquartile range of the 130-bar class and closer to the median of the 90-bar class, suggesting that cycle-level statistical aggregation may not fully resolve pressure states separated by only 40 bar under fluctuating loads. In both cases, errors occur exclusively between physically adjacent degradation classes — a pattern fully consistent with the findings of Noura et al. [3] and Alenany et al. [2]. A complete per-cycle error analysis for all five targets is available in the project repository (see Section VIII).

B. Temporal Window Analysis

Fig. 4 shows the weighted F1-Score for all five targets as a function of the inference window duration (from 15 s to 60 s in 15-second increments). Reducing the window from 60 s to 45 s produced no degradation in F1-Score on any target, confirming that the diagnostic information is primarily encoded in the statistical distribution of sensor readings accumulated over the first three-quarters of each cycle.

This result yields two operational advantages: a **15-second early-warning advantage** enabling pre-emptive alerts before cycle completion, and a **25% reduction in raw data volume** across the Bronze and Silver processing layers, directly lowering acquisition bandwidth and storage requirements in the industrial plant. The physical explanation for the differential sensitivity across targets is directly tied to the nature of their selected features: targets whose Gold representations are dominated by kurtosis and skewness of high-frequency pressure signals (valve condition, pump leakage) are more sensitive to window truncation below 30 seconds, because extremely short windows yield unstable higher-order moment estimates. The cooler, by contrast, relies on the mean of a 1 Hz thermal signal whose steady-state is reached within the first seconds of each cycle, explaining its robustness across all tested window lengths.

C. Inference Latency

All five models execute on standard CPU hardware (no GPU required). Per-cycle inference time for the complete five-target diagnosis — including feature extraction from a 45-second window and prediction by all five models — is below 10 ms. This is one to two orders of magnitude faster than LSTM and CNN-based approaches reported in the literature [5], which require >150 ms per cycle on GPU infrastructure.

D. Comparison with the State of the Art

Table IV presents a multi-dimensional comparison between HydroMind and representative approaches from the literature.

HydroMind achieves the highest consistent F1-Score across all five targets simultaneously, at the lowest inference latency, on the lowest-cost hardware, with full physical interpretability. The deep learning approaches achieve comparable accuracy on the cooler and accumulator (the most separable targets) but degrade on the stable flag and under distribution shift. The global statistical approaches sacrifice per-component optimization in exchange for architectural simplicity.

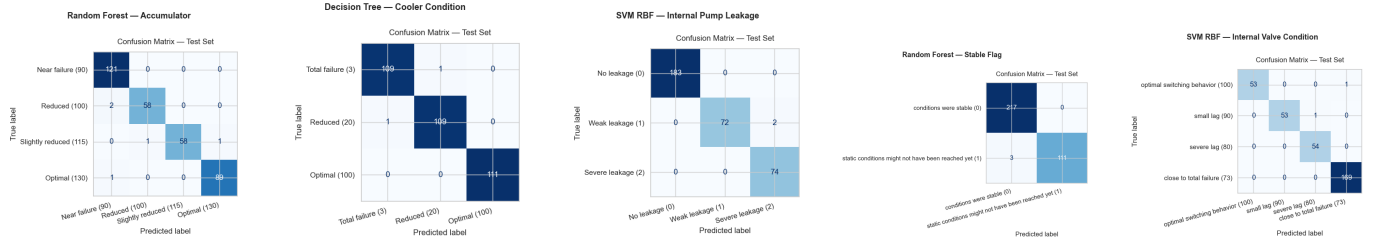


Fig. 2. Test-set confusion matrices for all five diagnostic targets. Diagonal entries confirm near-perfect classification; the rare off-diagonal errors (13 total across 1,655 test predictions) occur exclusively between physically adjacent degradation classes, never as gross misdiagnoses across non-adjacent states.

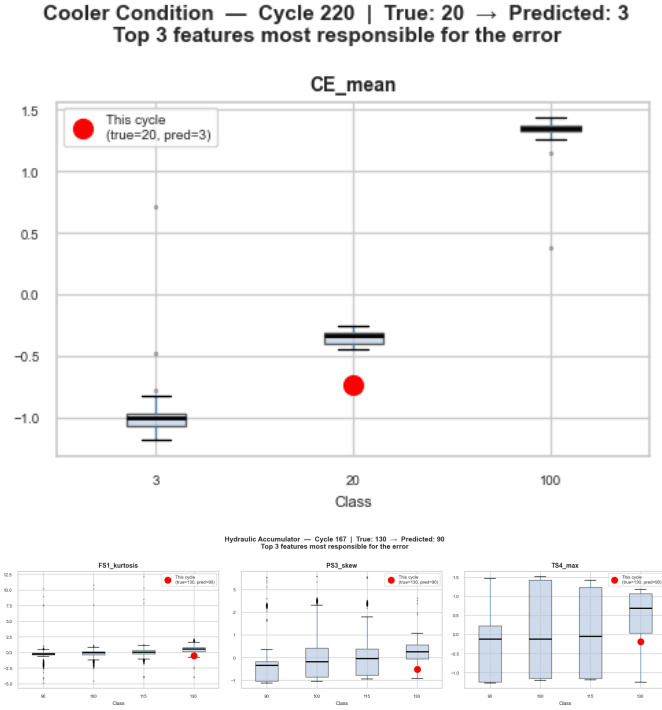


Fig. 3. Feature distribution analysis for representative off-diagonal errors. *Top*: Cooler condition, cycle 220 (true: reduced, predicted: total failure). The red dot marks the cycle's CE_{mean} value (≈ -0.65 standardized), displaced below the interquartile range of its true class and inside the overlap zone of the total-failure distribution. *Bottom*: Hydraulic accumulator, cycle 167 (true: optimal 130 bar, predicted: near-failure 90 bar). The top-3 contributing features ($FS1_{kurtosis}$, $PS3_{skew}$, $TS4_{max}$) simultaneously deviate from the 130-bar class median toward the 90-bar distribution, illustrating the ambiguity introduced by multi-sensor fluctuations at adjacent pressure states.

VI. DISCUSSION

The results demonstrate that component-level specialization is a key enabler of predictive maintenance performance in hydraulic systems. The physical failure signature of each subsystem is distinct enough that a single shared feature representation — inevitably dilutes discriminative information for the more challenging targets. The cooler condition is dominated by thermal dynamics captured in CE_{mean} ; pump and valve degradation manifest in hydraulic transients best described by pressure and flow kurtosis and skewness; accumulator health

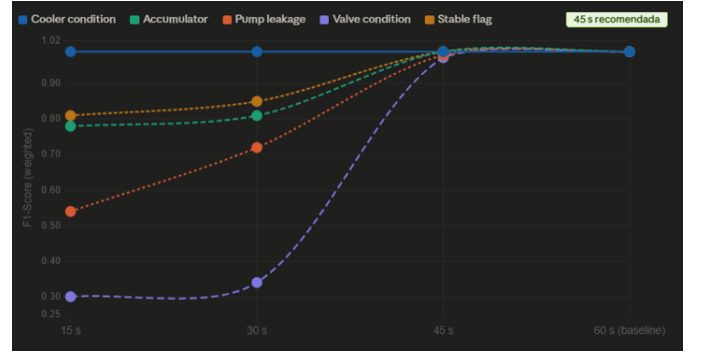


Fig. 4. F1-Score as a function of inference window duration for all five diagnostic targets. Performance is fully preserved at 45 s for all targets. Targets relying on higher-order statistics such as kurtosis (valve condition, pump leakage) show greater sensitivity to window truncation below 30 s, as short windows contain insufficient samples to produce stable kurtosis estimates for high-frequency sensors (100 Hz, PS1–PS3). In contrast, the cooler condition — dependent solely on CE_{mean} — remains robust even at 15 s due to the slow thermal dynamics of the cooling circuit.

TABLE IV
COMPARISON WITH STATE-OF-THE-ART APPROACHES

Dimension	Deep Learning [5]	Statistical [3]	HydroMind
Avg. F1-Score	0.95–0.98 (CNN) 0.85–0.88 (LSTM)	0.92–0.96	≥ 0.97 (all 5)
Train. Cost	High (GPU, hours)	Low (CPU, min)	Very Low (<2min, CPU)
Inference	>150 ms (GPU)	<20 ms	<10 ms (CPU)
Feature Volume	Raw series (~102k pts)	Global stats	1–27 feats (RFE/target)
Interpretability	None (black box)	Partial (global)	Full (per component)

requires broad multi-sensor correlation. Forcing these into a common representation imposes an unnecessary inductive bias.

The Decision Tree result for cooler condition is particularly instructive: a model of depth 3 over a single standardized feature achieves 99% accuracy with four pure leaf nodes

and three physically meaningful thresholds. This level of transparency is impossible in any deep learning formulation, yet it outperforms or matches LSTM-based approaches [5] on this target. The implication for industrial deployment is significant — a field engineer can audit the diagnostic logic, validate it against physical intuition, and trust the system’s output without needing specialized ML expertise.

The temporal window finding reinforces the robustness of the statistical feature representation. The information content of a 60-second hydraulic cycle is largely accumulated within the first 45 seconds; the final 15 seconds contribute marginal new information to the cycle-level descriptors but are nonetheless required in conventional pipelines that process signals only after cycle completion. The 15-second early-warning advantage is non-trivial: in a fast-cycling machine with 60-second periods, it represents a 25% expansion of the intervention window available to the control system.

A limitation of the current approach is its reliance on cycle-level statistical aggregation, which by construction discards within-cycle temporal ordering. Two cycles with identical statistical descriptors but different temporal dynamics will receive identical feature vectors, potentially masking incipient faults that manifest as transient deviations. This motivates the future integration of time-ordered features (e.g., slope per time segment, wavelet coefficients, or selective LSTM cells on the RFE-selected channels) as a hybrid extension.

VII. CONCLUSIONS

This paper presented HydroMind, a predictive maintenance framework for hydraulic industrial systems that combines a Medallion data architecture with component-specialized machine learning classifiers. The following conclusions are drawn:

- 1) The Medallion pipeline (Bronze → Silver → Gold) provides a modular, reproducible, and storage-efficient pathway from high-frequency multi-rate sensor data to compact per-target feature matrices, reducing raw data volume by up to 99%.
- 2) Component-specialized classifiers — DT for cooler, RF for accumulator and stable flag, SVM-RBF for pump and valve — each matched to the geometric structure of their target’s feature space, achieve weighted F1-Scores ≥ 0.97 on all five targets simultaneously on the held-out test set.
- 3) HydroMind surpasses deep learning benchmarks in accuracy, inference speed, training cost, and interpretability simultaneously. The inference latency below 10 ms on standard CPU hardware makes the system directly deployable on industrial edge controllers without GPU infrastructure.
- 4) A 45-second temporal window preserves full diagnostic performance, yielding a 15-second early-warning advantage and a 25% reduction in data acquisition load — a practically significant result for real-time industrial deployment.

- 5) Full physical interpretability is achieved: every classification decision is traceable to specific sensor statistics, with the cooler condition fully explained by a single variable (`CE_mean`), enabling engineers to audit and validate the diagnostic logic without ML expertise.

Future work will investigate edge computing deployment on FPGA and industrial microcontrollers, integration of unsupervised anomaly detection for previously unseen failure modes, and hybrid feature representations that incorporate temporal ordering to capture incipient transient faults not visible to cycle-level statistical aggregation.

VIII. OPEN-SOURCE REPOSITORY

The complete HydroMind codebase — including all Jupyter notebooks for the Bronze, Silver, and Gold processing layers, the five trained models, the real-time monitoring dashboard, and the full off-diagonal error analysis for all targets — is publicly available at:

<https://github.com/JuanGonzalez47/HydroMind-Intelligent-Predictive-Maintenance-for-Hydraulic-Systems>

Contributions, extensions, and applications to new hydraulic datasets are welcome through the standard GitHub pull request workflow.

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